Hyperspectral Pansharpening Based on Improved Deep Image Prior and Residual Reconstruction

Wele Gedara Chaminda Bandara, *Student Member, IEEE*,
Jeya Maria Jose Valanarasu, *Student Member, IEEE*,
and Vishal M. Patel, *Senior Member, IEEE*

***Abstract*— Hyperspectral pansharpening aims to synthesize a low-resolution hyperspectral image (LR-HSI) with a registered panchromatic (PAN) image to generate an enhanced HSI with high spectral and spatial resolution. Recently, the proposed HS pansharpening methods have obtained remarkable results using deep convolutional networks (ConvNets), which typically consist of three steps: 1) upsampling the LR-HSI; 2) predicting the residual image via a ConvNet; and 3) obtaining the final fused HSI by adding the outputs from first and second steps. Recent methods have leveraged deep image prior (DIP) to upsample the LR-HSI due to its excellent ability to preserve both spatial and spectral information, without learning from large datasets. However, we observed that the quality of upsampled HSIs can be further improved by introducing an additional spatial-domain constraint to the conventional spectral-domain energy function. We define our spatial-domain constraint as the L_1 distance between the predicted PAN image and the actual PAN image. To estimate the PAN image of the upsampled HSI, we also propose a learnable spectral response function (SRF). Moreover, we noticed that the residual image between the upsampled HSI and the reference HSI mainly consists of edge information and very fine structures. In order to accurately estimate fine information, we propose a novel overcomplete network, called HyperKite, which focuses on learning high-level features by constraining the receptive from increasing in the deep layers. We perform experiments on three semisynthetic and one real HSI datasets to demonstrate the superiority of our DIP-HyperKite over the state-of-the-art pansharpening methods. The deployment codes, pretrained models, and final fusion outputs of our DIP-HyperKite and the methods used for the comparisons will be publicly made available at https://github.com/wgcban/DIP-HyperKite.git.**

***Index Terms*— Deep image prior (DIP), hyperspectral image fusion, hyperspectral pansharpening, overcomplete representations, spatial and spectral constraints.**

I. Introduction

HYPERSPECTRAL images (HSIs) with a large number of spectral bands have gained immense attention in the field of remote sensing due to its applications in broad research areas such as classification [1], unmixing [2], anomaly detection [3], change detection [4], etc. However, due to the limited incident energy available when capturing an image, hyperspectral imaging systems face tradeoffs between spectral resolution, spatial resolution, and signal-to-noise ratio (SNR) [5]. For this reason, hyperspectral imaging systems can provide images with high spectral resolution but with low spatial resolution. In contrast, multispectral imaging systems can provide data with high spatial resolution but with fewer spectral bands (e.g., panchromatic images or multispectral images (MSIs) with three or four spectral bands). Low spatial resolution in HSIs leads to relatively poor performance in some practical remote sensing applications, such as road topology extraction [6], and spectral unmixing [7]. Therefore, full-resolution HSIs with high spatial and spectral resolution are desired. One way to obtain such ideal HSIs is to fuse high spectral resolution HSIs with high spatial resolution PAN/MSIs. This fusion process is called HS pansharpening in the remote sensing literature, which is indeed a form of super-resolution [8].

Traditional pansharpening methods can be mainly divided into five classes [5], [9]: 1) Component substitution (CS); 2) multiresolution analysis (MRA); 3) Bayesian; 4) matrix factorization; and 5) variational. CS methods rely on substituting the spatial component of the HSI with the MSI/PAN image. The family of CS contains algorithms such as Gram–Schmidt adaptive (GSA) [10], [11], principal component analysis (PCA) [12]–[14], and intensity-hue-saturation (IHS) [15]. Even though the CS methods usually generate pansharpened HSIs with accurate spatial information, sometimes they suffer from critical spectral distortions. The MRA approaches are based on injecting the spatial details obtained through the multiscale decomposition of the MSI/PAN image into the HSI. In order to extract the spatial details from the PAN image, several algorithms have been proposed in the literature, such as decimated wavelet transform (DWT) [16], undecimated wavelet transform (UDWT) [17], smoothing filter-based intensity modulation (SFIM) [18], modulation transfer function with generalized Laplacian pyramid (MTF-GLP) [19], and MTF-GLP with high-pass modulation (MTF-GLP-HPM) [20]. In contrast to the CS methods, the MRA family performs better in spectral preservation but is more sensitive to registration errors, which may cause critical distortions in the spatial domain. Due to these inherent advantages and disadvantages of CS and MRA approaches, there have been works that attempted to combine both CS and MRA methods. One of the

Manuscript received July 5, 2021; revised September 2, 2021 and November 30, 2021; accepted December 25, 2021. Date of publication December 28, 2021; date of current version March 3, 2022. This work was supported by the NSF CAREER Award under Grant 2045489. *(Corresponding author: Wele Gedara Chaminda Bandara.)*

The authors are with the Whiting School of Engineering, The Johns Hopkins University, Baltimore, MD 21218 USA (e-mail: wbandar1@jhu.edu; jvalana1@jhu.edu; vpatel36@jhu.edu).

Digital Object Identifier 10.1109/TGRS.2021.3139292

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

representatives of hybrid CS and MRA algorithm is guided filter PCA (GFPCA) [21]. The Bayesian-based methods also provide a convenient way to regularize the fusion methods by modeling the posterior distribution of the target HSI provided by the LR-HSI and MSI/PAN image. Examples of the algorithms based on the Bayesian inference framework include convex regularization under a Bayesian framework (abbreviated as Hysure) [22], naive Bayesian Gaussian prior (abbreviated as BF) [23], and sparsity promoted Gaussian prior (abbreviated as BFS) [24]. Finally, the coupled nonnegative matrix factorization (abbreviated as CNMF) is one of the examples for matrix factorization-based methods, which regularizes the fusion problem by using the priors of spectral unmixing [25]. More recently, the variational methods have gained a significant attention in HS pansharpening [26]–[28]. These methods perform pansharpening by modeling the relationship between PAN, LR-HSI, and HR-HSI images into an objective function based on some prior knowledge or certain assumptions. However, the fusion performance of traditional pansharpening approaches is generally limited due to their inadequate representation ability [29]. In addition, the algorithms mentioned above may result in severe quality degradation when the assumptions do not align with a particular dataset. Furthermore, most traditional pansharpening approaches typically reach the optimal solution through an iterative process, which is time-consuming and inefficient.

Recently, deep learning (DL) models based on convolutional neural networks (ConvNets) have also been introduced for the HS pansharpening problem due to ConvNets' excellent ability to learn high-level features automatically [8], [29]. ConvNet-based HS pansharpening methods generally consist of three steps.

1) *Upsampling Step:* Upsampling the LR-HSI to the spatial resolution of the PAN image.
2) *Residual Reconstruction Step:* Concatenating the upsampled HSI and PAN image along the spectral dimension and passing it through a residual learning network to learn the residual image.
3) *Final Fusion Step:* Obtaining the final fused HSI by adding the upsampled HSI and the residual image.

There have been many methods proposed to upsample LR-HSI to the spatial resolution of PAN. In the earliest studies, nearest neighbor and bicubic interpolation were the famous methods to perform upsampling. However, the methods mentioned above conduct upsampling on each band of the LR-HSI successively, thus ignoring the high spectral correlation of HSIs, which may lead to spectral distortions [30], [31]. In order to minimize the spectral distortion, data-driven upsampling techniques (i.e., deep super-resolution networks) have also been utilized in HS pansharpening. The LapSRN [32] network is an example of such a data-driven super-resolution method, which progressively super-resolves an LR image in a coarse-to-fine manner in a Laplacian pyramid framework. However, the LapSRN method requires a large number of images for training, which is impractical in the HS domain due to the limited number of datasets available to the public. A remedy to the problem mentioned above was proposed by Lempitsky *et al.* [31] where they proposed a DL-based super-resolution framework called deep image prior (DIP). The proposed method uses a randomly initialized ConvNet to upsample an image, using its structure as an image prior, similar to bicubic upsampling. However, this method does not require any training but produces much cleaner results with sharper edges. Motivated by the super-resolution performance of DIP in the RGB domain, researchers have applied DIP to the HS pansharpening problem [30], [33] and achieved impressive results. However, we observed that the energy function defined in HS DIP upsampling directly applies the energy function formulated for the RGB DIP process, where they only impose spectral-domain constraint by computing the L_1 distance between the downsampled version of the target upsampled HSI and the LR-HSI. However, the existing HS DIP methods do not impose any spatial-domain constraint by utilizing the available PAN image. We address this issue by introducing an additional spatial-domain constraint to the HS DIP process as our first contribution.

For residual reconstruction, various ConvNet architectures have been proposed in the literature to accurately predict the residual component between the upsampled HSI and the reference HSI with less spectral and spatial distortion. Among those, Masi *et al.* [34] were the first to introduce simple three-layer ConvNet architecture for the residual learning. Furthermore, He *et al.* [35] improved the spatial and spectral prediction capability of Giuseppe's work (abbreviated as HyperPNN) by introducing spectral and spatial prediction modules. To further enhance the representational power of ConvNets, attention mechanisms [36] have also been introduced. Among those, Zheng *et al.* [30] proposed a spatial and spectral attention mechanism (abbreviated as DHP-DARN) for the residual learning in which they cascade several channel-spatial-attention residual blocks to adaptively learn more informative channel-wise and spatial-domain features simultaneously. More recently, Xu *et al.* [37] proposed a design (abbreviated as SDPNet) based on two encoder–decoder networks to extract deep-level features from two types of source images with densely connected blocks to strengthen feature propagation. However, we experimentally observed that most of the existing residual learning methods fail when predicting the high-frequency information, such as edges and delicate structures in the residual image. The main reason for this observation is due to the increasing receptive field of the network in the deep layers. Motivated by this observation, we introduce an overcomplete network, called HyperKite, for residual reconstruction task as our second contribution, which constrains the receptive field from increasing in deep layers, thus extracting more high-frequency information.

The main contributions of this article are summarized as follows.

1) A novel spatial constraint is introduced for the DIP upsampling process. To the best of our knowledge, this is the first study that integrates both spatial and spectral-domain constraints to the DIP upsampling. The proposed spatial constraint significantly improves the spatial and spectral performance measures of the upsampled HSIs.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

2) An overcomplete network, called HyperKite, is proposed for the residual reconstruction, which is highly capable of extracting high-frequency information of the residual image by appropriately constraining the receptive field of the network.
3) We conduct extensive experiments to clearly demonstrate the improvements brought in from our contributions to the HS pansharpening. We compared the fusion performance of DIP-HyperKite with both conventional and DL-based approaches. The deployment codes, pretrained models, and final fusion results of our DIP-HyperKite as well as the comparison methods in the results and discussion will be publicly made available at https://github.com/wgcban/DIP-HyperKite.git.

The rest of this article is organized as follows. Section II provides some basics of DIP and overcomplete representations. In Section III, the proposed DIP-HyperKite is described in detail. Section IV describes the datasets and performance metrics that we used in the experiments. In Section V, the experimental results on different datasets are presented. Finally, the conclusions are drawn in Section VI.

II. Related Work

A. DIP for HSI Upsampling

Generally, ConvNets have an excellent ability to learn realistic image priors from a large amount of visual data, placing them in leading positions on the benchmarks of various image processing tasks [38], [39]. Contrary to the general opinion on deep networks that they require large data to capture image priors, DIP [31] has shown that a randomly initialized network can capture low-level image statistics before any training. Concretely, in HS pansharpening, DIP can generate the upsampled HSI $\mathbf{x}_{\text{dip}}$ of the LR-HSI $\mathbf{y}$ with spatial upsampling factor β by taking a fixed randomly initialized vector $\mathbf{z}$ as the input and utilizing the deep network as a parametric function $\mathbf{x}_{\text{dip}} = f_\theta(\mathbf{z})$. Next, the network is optimized over its parameters θ to obtain the upsampled HSI $\mathbf{x}_{\text{dip}}$ as follows:

$$\mathbf{x}_{\text{dip}} = \min_{\mathbf{x}_{\text{dip}}} Q\left(\mathbf{x}_{\text{dip}}; \mathbf{y}\right) + R\left(\mathbf{x}_{\text{dip}}\right) \tag{1}$$

where $Q(\mathbf{x}_{\text{dip}}; \mathbf{y})$ is an energy function that controls the fidelity toward the LR-HSI $\mathbf{y}$, and $R(\mathbf{x}_{\text{dip}})$ is a regularization function based on prior knowledge. In [31], it has been shown that the regularization term $R(\mathbf{x}_{\text{dip}})$ can be implicitly substituted by the deep network. Therefore, the minimization problem in (1) has simplified to optimizing the network over its parameters θ as follows:

$$\theta^* = \arg\min_{\theta} Q\left(\mathbf{x}_{\text{dip}}; \mathbf{y}\right) \quad \text{s.t. } \mathbf{x}_{\text{dip}} = f_\theta(\mathbf{z}) \tag{2}$$

where θ^* denotes the optimal set of parameters of the network. Furthermore, the most straightforward and commonly utilized energy function in HS pansharpening is the L_1 distance [35] between the downsampled version of the upsampled HSI $\mathbf{x}_{\text{dip}}$ and the LR-HSI $\mathbf{y}$ as follows:

$$Q\left(\mathbf{x}_{\text{dip}}; \mathbf{y}\right) = \left\| d\left(\mathbf{x}_{\text{dip}}\right) - \mathbf{y} \right\|_1 \quad \text{s.t. } \mathbf{x}_{\text{dip}} = f_\theta(\mathbf{z}) \tag{3}$$

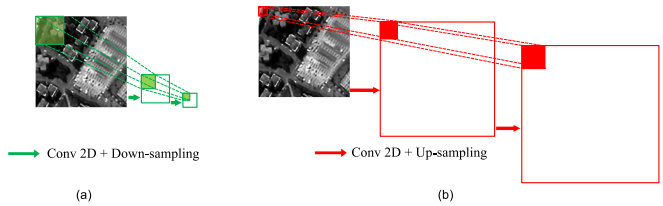


Fig. 1. (a) Effect of undercomplete ConvNet on receptive field where the deeper layers focus on a larger region of the input thus extracting high-level/low-frequency information. (b) Effect of overcomplete ConvNet on receptive field where the deeper layers focus on a much smaller region in the input, thus extracting low-level/high-frequency information.

where $d(\cdot)$ denotes the downsampling operator by a factor of β.

B. Overcomplete ConvNets

Most of the current architectures in DL are "encoder–decoder"-based [40]–[42]. Here, the encoder translates the high-dimensional input to a low-dimensional latent space, whereas the decoder learns to take the latent low-dimensional representation back to a high-dimensional output. These types of architectures learn low-level features at their initial layers and high-level features at their deeper layers. These are termed undercomplete networks as the input is taken to a lower spatial dimension in the latent space.

In signal processing, overcomplete dictionaries are widely used for their highly robust characteristic [43]. The number of basis functions here is more than the number of input signal samples, which enables higher flexibility for capturing structure in data. In [44], overcomplete autoencoders were found to be better feature extractors for denoising when compared to undercomplete autoencoders. In an overcomplete network [45], the encoder takes the input data to a higher spatial dimension unlike a traditional encoder. This is achieved by using an upsampling layer after every convolutional layer in the encoder. Using upsampling layers in the encoder causes the receptive field to be constrained in the deep layers. This causes the deep layers in the network to learn more fine-context high-frequency information when compared to undercomplete networks. An increase in receptive field for an overcomplete network can be generalized in an ith layer as follows:

$$RF(\text{w.r.t } I) = \left(\frac{1}{2}\right)^{2(i-1)} \times k \times k \tag{4}$$

where the initial receptive field of the conv filter is assumed to be $k \times k$ on the image I. This phenomenon has been visualized in Fig 1. As shown in Fig. 1(b), by employing an upsampling layer after every convolutional layer in the encoder, the overcomplete network restricts the receptive field size to a smaller region, which forces the network to learn very fine edges as it tries to focus heavily on smaller regions. This is completely different from the conventional undercomplete architectures where they perform downsampling after each convolution block, which makes the network to focus on a much larger region in the input, as shown in Fig. 1(a).

Overcomplete networks in DL is a new topic and was initially proposed for medical image segmentation of small

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

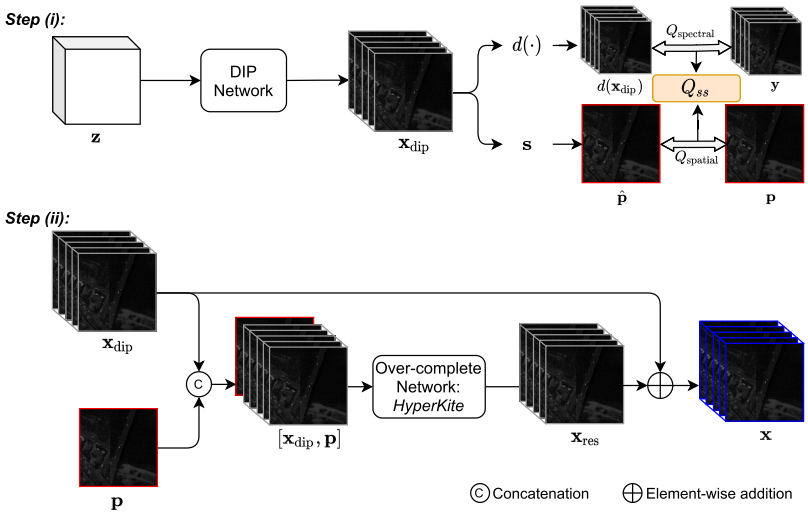


Fig. 2. Overall flowchart of our proposed DIP-HyperKite for HS pansharpening. In the first step, we upsample the LR-HSI $\mathbf{y}$ via DIP process to obtain the upsampled HSI $\mathbf{x}_{\text{dip}}$. The DIP process takes a fixed noise tensor $\mathbf{z}$ as an input for a given LR-HSI $\mathbf{y}$ and produces the upsampled HSI $\mathbf{x}_{\text{dip}}$ by optimizing the proposed spatial + spectral energy function Q_{ss} over the DIP network parameters θ. In the second step, we take the upsampled HSI $\mathbf{x}_{\text{dip}}$ and the PAN image $\mathbf{p}$ as inputs to predict the residual component $\mathbf{x}_{\text{res}}$ using our proposed overcomplete network—HyperKite. Finally, the predicted residual image $\mathbf{x}_{\text{res}}$ is added to the upsampled HSI $\mathbf{x}_{\text{dip}}$ to obtain the pansharpen HSI $\mathbf{x}$.

anatomy [45]. It has since been successfully extended to solve fine-context requiring tasks, such as fine edge segmentation of 3-D volumes [46], deep subspace clustering [47], MRI reconstruction [48], adversarial defense against videos [49], and image restoration problems like single image deraining [50].

III. Methodology

The overall flowchart of the proposed DIP-HyperKite for HS pansharpening is shown in Fig. 2. As can be seen from Fig. 2, the proposed method consists of two main steps. In the first step, the LR-HSI $\mathbf{y} \in \mathbb{R}^{l \times w \times h}$ with $w \times h$ pixels and l spectral bands is upsampled to the spatial resolution of the PAN image $\mathbf{p} \in \mathbb{R}^{1 \times \beta w \times \beta h}$, where β denotes the ratio between spatial resolution of $\mathbf{p}$ and $\mathbf{y}$. We denote the output from the DIP process as $\mathbf{x}_{\text{dip}} \in \mathbb{R}^{l \times \beta w \times \beta h}$. In the second step, we train an overcomplete deep network, which takes upsampled HSI $\mathbf{x}_{\text{dip}}$ and the corresponding PAN images $\mathbf{p}$ as inputs to predict the residual component $\mathbf{x}_{\text{res}}$ between the upsampled HSI $\mathbf{x}_{\text{dip}}$ and the reference HSI $\mathbf{x}_{\text{ref}}$.

A. Upsampling via DIP

As shown in Fig. 2, the low-resolution HSI $\mathbf{y}$ is upsampled to the spatial resolution of the PAN image $\mathbf{p}$ using the DIP. This recently introduced DIP method is different from the other existing upsampling techniques, such as bicubic interpolation and LapSRN [32]. The main advantage of DIP over these conventional methods is that it does not require a large dataset for training. In other words, for each LR image $\mathbf{y}$, the DIP network takes a fixed random tensor $\mathbf{z}$ as an input and optimizes the network parameters θ by minimizing the loss function Q that is defined in terms of the output upsampled image $\mathbf{x}_{\text{dip}}$ and available LR-HSI $\mathbf{y}$ as given in (3). In contrast, the LapSRN network utilized in [51] is highly relied upon the RGB image datasets and the knowledge adaptation techniques. Furthermore, the bicubic and LapSRN methods upsample each band in the HSI separately, thus ignoring the high spatial correlation between the spectral bands, which results in the loss of spatial details. Although the DIP method is capable of producing high-quality upsampling images compared to the other existing methods, it only utilizes the information from the LR-HSI $\mathbf{y}$, thus only imposing constraint on the spectral domain. However, we observed that the quality of the sampled HSIs can be further improved by incorporating an additional spatial constraint in the loss function using the available PAN image $\mathbf{p}$. In the next section, we explain our novel spatial + spectral loss function.

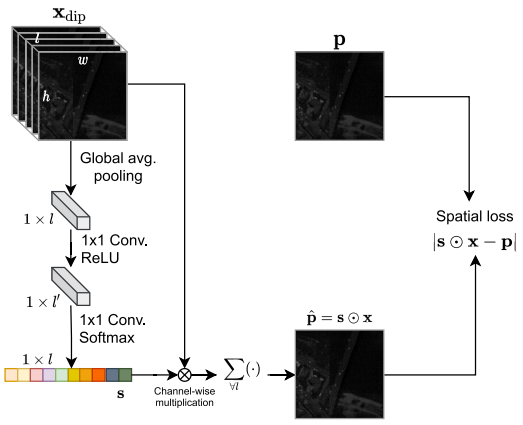


Fig. 3. Proposed learnable SRF $\mathbf{s}$ and the computational procedure of evaluating the spatial loss term Q_{spatial}. We take the upsampled HSI $\mathbf{x}_{\text{dip}}$ as the input and feed it in to a GAP layer, which yields a vector with a single entry for each spectral band. Then, we pass it through a gating mechanism by forming a bottleneck with two FC layers (1×1 convolutions) around the nonlinearity to learn the spectral response of each band. Next, we apply a softmax activation function to obtain normalized spectral response $\mathbf{s}$ and then take the channel-wise multiplication followed by channel averaging to obtain the estimated PAN image $\hat{\mathbf{p}}$. Finally, we compute the L_1 distance between the estimated PAN image $\hat{\mathbf{p}}$ and the reference PAN image $\mathbf{p}$ to obtain the spatial loss Q_{spatial}.

1) Proposed Spatial + Spectral Energy Function for HS DIP: As we discussed in Section II-A, the energy function

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

given in (3) enforces a constraint only in spectral domain by defining the L_1 distance between upsampled HSI $\mathbf{x}_{\text{dip}}$ and the LR HSI $\mathbf{y}$. Instead, we propose a loss function (denoted by Q_{ss}) for HS DIP, which enforces the constraints in both spatial and spectral domains as follows:

$$Q_{ss} = \underbrace{\left\| d\left(\mathbf{x}_{\text{dip}}\right) - \mathbf{y} \right\|_1}_{\text{spectral energy}} + \lambda \underbrace{\left\| \left(\sum_{i \in \forall l} \mathbf{s}[i] \odot \mathbf{x}_{\text{dip}}[i] \right) - \mathbf{p} \right\|_1}_{\text{spatial energy}} \quad (5)$$

where $\mathbf{s} \in \mathbb{R}^{1\times l}$ denotes the spectral response function (SRF), $\mathbf{s}[i]$ (scalar) is the spectral response of the ith band, $\mathbf{x}[i] \in \mathbb{R}^{h\times w}$ is the ith band image of the upsampled HSI $\mathbf{x}_{\text{dip}}$, $\odot$ is the element-wise multiplication, and λ is a regularization constant. The first term in (5) enforces the spectral constraint on $\mathbf{x}_{\text{dip}}$ as in (3), and the additional second term enforces the constraint in spatial domain on $\mathbf{x}_{\text{dip}}$ by utilizing the available PAN image $\mathbf{p}$.

In the simplest case, the SRF can be approximated as the average across all spectral bands (i.e., $\mathbf{s}[i] = 1/l; \forall i \in [1, l]$) [22], [52]. In this scenario, the spatial loss term in (5) enforces that the average across all the spectral bands in upsampled HSI $\mathbf{x}_{\text{dip}}$ to be close as possible to the PAN image $\mathbf{p}$, thus assuming a flat (i.e., uniform) spectral response. However, in general, this assumption is not valid as spectral response varies with wavelength coverage and different spectral bands describe the same semantic information across a wide spectral range with varying quality (i.e., PSNR) [53].

A recent attempt [53] estimates the SRF $\mathbf{s}$ by utilizing the larger eigenvalue of the structure tensor (ST) matrix (originally proposed in the Harris corner detection algorithm [54]). However, this method cannot be directly utilized in an end-to-end DL network due to the difficulties encountered while performing backpropagation. In addition, it is highly computationally complex as it requires to compute derivatives of each band image along both the x- and y-directions at each iteration of learning as part of constructing the structure tensor matrix. Instead, we propose a computationally lightweight and learnable SRF, which can be easily integrated into the spatial loss term in (5) and can be simultaneously learned with DIP.

In this section, we describe our novel way of estimating the SRF that is computationally lightweight and differentiable and can be easily integrated into the existing DIP learning process. The overall computational procedure of estimating the SRF and thereby evaluating the spatial energy that we introduced for the DIP process in (5) is graphically shown in Fig. 3. First, we assume that the spectral response is proportional to the ratio of information in each spectral band. The next problem that arises with this assumption is how do we quantify the information embedded in each spectral band. Motivated by recently proposed squeeze-and-excitation networks, we utilize global average pooling to quantify the global information present in each band. Formally, a statistic $\mathbf{q} \in \mathbb{R}^{1\times l}$, which quantifies the informative features in each spectral band, is generated by shrinking the upsampled HSI $\mathbf{x}_{\text{dip}}$ through its spatial dimensions $h \times w$ such that the ith element in $\mathbf{q}$ is calculated as

$$\mathbf{q}(i) = \frac{1}{h \times w} \sum_{\tilde{h}=1}^{h} \sum_{\tilde{w}=1}^{w} \mathbf{x}_i\left(\tilde{w}, \tilde{h}\right). \quad (6)$$

Next, we use a simple gating mechanism to capture the dependencies among spectral bands using the band-wise descriptor $\mathbf{q}$ that we obtained in the previous step. The gating mechanism consists of two fully connected (FC) layers that give the network more flexibility to automatically learn the best SRF during the training process, resulting in better performance than direct normalization that does not involve any parameters to learn. We parameterize the SRF $\mathbf{s}$ by forming a bottleneck with two FC layers around the nonlinearity as follows:

$$\mathbf{s} = \sigma\left(\mathbf{w}_2\, \delta(\mathbf{w}_1 \mathbf{q})\right) \quad (7)$$

where σ is the sigmoid activation function, δ is the ReLU nonlinearity, and $\mathbf{w}_1$ and $\mathbf{w}_2$ are the learnable weight matrices. Here, we use sigmoid activation to guarantee that the spectral responses of all the bands sump up to one.

2) DIP Network: Fig. 4 shows a U-Net like deep network that we used for the DIP method. The DIP network includes five downsampling blocks $\mathbf{d}[i]$, five upsampling blocks $\mathbf{u}[i]$, and five skip-connection blocks $\mathbf{sk}[i]$ ($i = 1, 2, .., 5$). We use stride convolutions as the downsampling operator, bilinear upsampling as the upsampling operator, and Lanczos2 as nonlinearity. We initialize the input noise vector with uniform noise between 0 and 0.1. Table I tabulates the values of all the hyperparameters of DIP network.

TABLE I
HYPERPARAMETER VALUES OF THE DIP NETWORK

Hyperparameter	Value
z	$\mathbb{R}^{32\times\beta h\times\beta w} \sim U(0, 0.1)$
$n_d = n_u$	$[128, 128, 128, 128, 128]$
$k_d = k_u$	$[3, 3, 3, 3, 3]$
n_s	$[4, 4, 4, 4, 4]$
k_s	$[1, 1, 1, 1, 1]$
Optimizer	Adam
Number of iterations	1300
Learning rate	0.001
Weight decay	0.0001
Momentum	0.9
Batch size	4
LeakyReLU slope	0.2

TABLE II
HYPERPARAMETER VALUES OF HYPERKITE

Hyperparameter	Value
n	$[32, 64, 128, 128, 64, 32, l]$
k	$[3, 3, 3, 3, 3, 3, 3]$
Optimizer	Adam
Num_it	2500
Learning rate	0.001
Weight decay	0.0001
Momentum	0.9
Batch size	4
LeakyReLU slope	0.2

B. Residual Learning via Overcomplete HyperKite

Our motivation to design an overcomplete network for the residual learning task emerged after observing the residual images between DIP upsampled image $\mathbf{x}_{\text{dip}}$ and reference HSI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

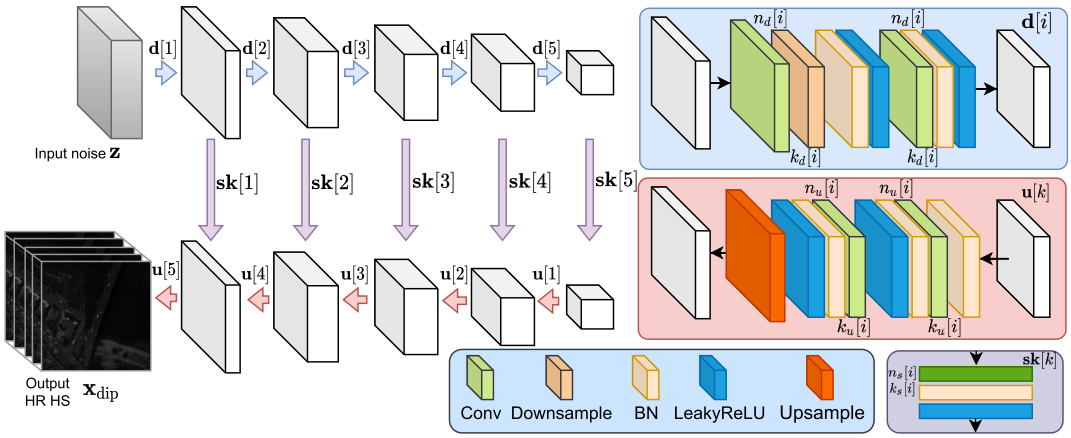


Fig. 4. DIP network utilized for the upsampling process. The DIP network is a U-Net like network that consists of five downsampling blocks $\mathbf{d}[i]$, five upsampling blocks $\mathbf{u}[i]$, and five skip-connection blocks $\mathbf{sk}[i]$ $(i = 1, 2, .., 5)$. The values of all the hyperparameters of DIP network are summarized in Table I.

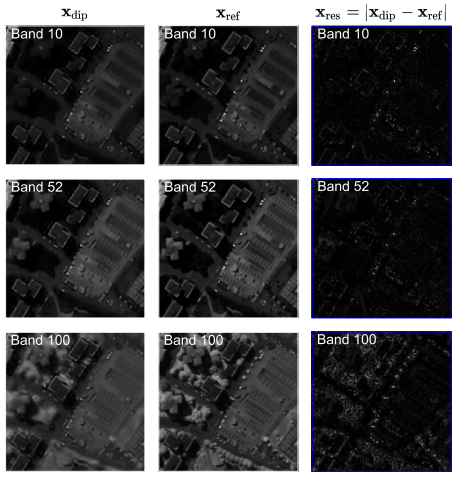


Fig. 5. We observed that the residual component $\mathbf{x_{res}}$ (see the third column) between upsampled HSI $\mathbf{x}_{\text{dip}}$ (see the first column) and the reference HSI $\mathbf{x_{ref}}$ (see the second column) mainly consists of boundary information and very fine structures. To support this observation, we show the residual component $\mathbf{x_{res}}$ for three different wavelength bands (i.e., bands 10, 52, and 100) in the Pavia Center dataset, which will be introduced in Section IV-A. This observation motivated us to use an overcomplete network for the residual learning task, which is highly capable of learning low-level features such as fine edges and structures by transforming the input image into a higher dimension. We recommend that readers zoomed-in view on this image to get a close-up view.

$\mathbf{x}_{\text{ref}}$, as visualized in Fig. 5. As we can see from Fig. 5, the residual images correspond to different wavelength bands, which mainly consists of boundary information such as edges and other high-frequency components. In order to accurately capture this fine information, we design an overcomplete HyperKite for the residual learning, as shown in Fig. 6.

The proposed HyperKite consists of an initial feature extraction network (IFEN), a high-dimensional feature mapping network (HDFMN), and a final residual reconstruction network (FRRN). The input to the HyperKite $\mathbf{x}_{\text{in}}$ is obtained by concatenating the upsampled HSI $\mathbf{x}_{\text{dip}}$ and the PAN image $\mathbf{p}$ along the spectral dimension (denoted as $[\mathbf{x}_{\text{dip}}, \mathbf{p}]$). The HyperKite starts with the IFEN layer, where one 3×3 convolutional layer is applied followed by batch normalization (BN) and LeakyReLU nonlinearity to extract the initial feature representation as

$$\mathbf{F}_{\mathcal{D}_1} = f_{\text{IFEN}}(\mathbf{x}_{\text{in}}) \tag{8}$$

where $f_{\text{IFEN}}(\cdot)$ denotes the 3×3 convolution followed by LeakyReLU and batch normalization, $\mathbf{F}_{\mathcal{D}_1}$ denotes the extracted features transformed from $\mathbf{x}_{\text{in}}$ in $\mathcal{D}_1 \in \mathbb{R}^{n[0] \times \beta w \times \beta h}$ dimensional pixel space, and $n[0]$ is the number of filters in the convolutional layer. Fig. 7(a) shows six example feature maps of $\mathbf{F}_{\mathcal{D}_1}$ for the 20th patch of the Pavia Center dataset that we will be introduced in Section IV. As we can see from the figure, the initial feature extraction network $f_{\text{IFEN}}(\cdot)$ extracts a low-level feature of the input $\mathbf{x}_{\text{in}}$. In order to capture high-level features that required for the residual learning, we successively transform the output of IFEN into three higher dimensional pixel spaces by utilizing the "bilinear" upsampling denoted as $\mathcal{D}_2 \in \mathbb{R}^{n[2] \times 2\beta w \times 2\beta h}$, $\mathcal{D}_4 \in \mathbb{R}^{n[3] \times 4\beta w \times 4\beta h}$, and $\mathcal{D}_8 \in \mathbb{R}^{n[4] \times 8\beta w \times 8\beta h}$. Then, we perform 3×3 convolution followed by BN and LeakyReLU to extract meaningful high-level features at each higher dimensional space as

$$\mathbf{F}_{\mathcal{D}_2} = f_{\mathcal{D}_2}\left(\uparrow \mathbf{F}_{\mathcal{D}_1}\right) \tag{9}$$

$$\mathbf{F}_{\mathcal{D}_4} = f_{\mathcal{D}_4}\left(\uparrow \mathbf{F}_{\mathcal{D}_2}\right) \tag{10}$$

$$\mathbf{F}_{\mathcal{D}_8} = f_{\mathcal{D}_8}\left(\uparrow \mathbf{F}_{\mathcal{D}_4}\right) \tag{11}$$

where $\uparrow$ denotes the "bilinear" interpolation by a factor of 2 and $f_{\mathcal{D}_{\text{d}}}(\cdot) : \text{d} \in \{2, 4, 8\}$ denotes the 3×3 convolution layer followed by BN and LeakyReLU at the dth higher dimensional feature space. Next, we successively transform the extracted high-level features to the original dimensional space $\mathcal{D}_1$ by employing "bilinear" downsampling and skip connections. Formally, we can define the operations of HDFMN as

$$\widetilde{\mathbf{F}}_{\mathcal{D}_4} = \widetilde{f}_{\mathcal{D}_4}(\downarrow \mathbf{F}_8 \oplus \mathbf{F}_4) \tag{12}$$

$$\widetilde{\mathbf{F}}_{\mathcal{D}_2} = \widetilde{f}_{\mathcal{D}_2}\left(\downarrow \widetilde{\mathbf{F}}_{\mathcal{D}_4} \oplus \mathbf{F}_2\right) \tag{13}$$

$$\widetilde{\mathbf{F}}_{\mathcal{D}_1} = \downarrow \widetilde{\mathbf{F}}_{\mathcal{D}_2} \tag{14}$$

where $\downarrow$ denotes the "bilinear" downsampling by a factor of 2, $\oplus$ denotes the feature concatenation operator, $f_{\mathcal{D}_{\text{d}}}(\cdot) : \text{d} \in \{0, 2, 4\}$ denotes the 3×3 convolution followed by BN and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

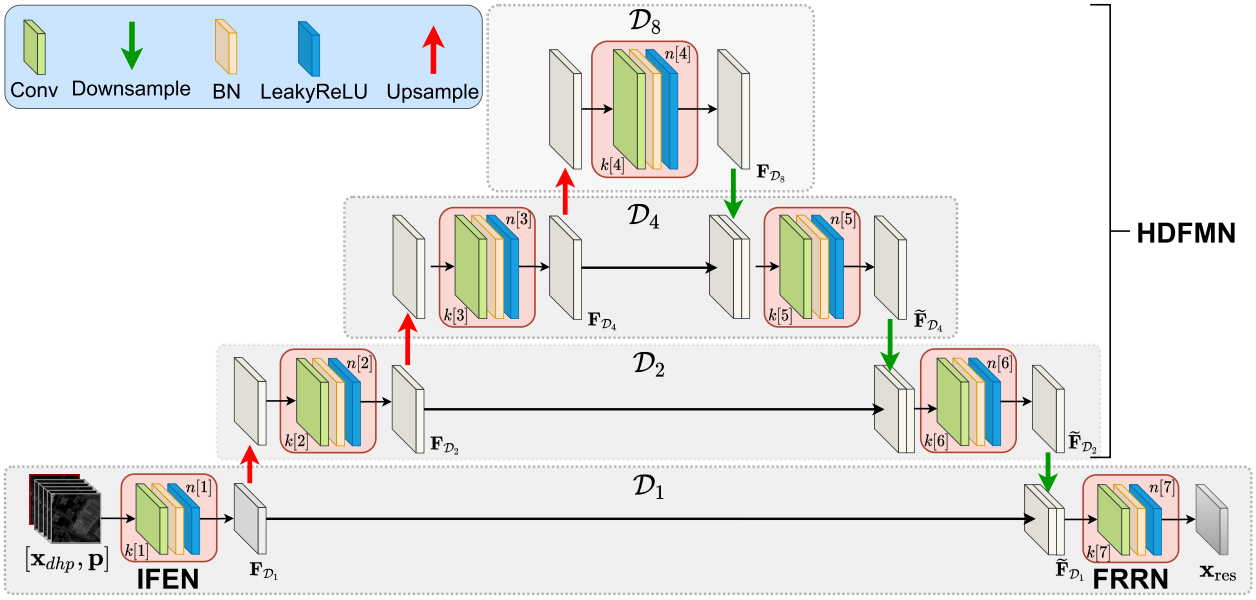


Fig. 6. Proposed HyperKite architecture for the residual prediction task. We denote the kernel size and the number of filters associated with each convolution block (shown in red color box) as $k[\cdot]$ and $n[\cdot]$, respectively. The values of all hyperparameters for HyperKite are summarized in Table II.

LeakyReLU at the dth dimensional feature space, and $\widetilde{\mathbf{F}}_{\mathcal{D}_\mathrm{d}}$ is the most relevant high-level features obtained at $\mathcal{D}_\mathrm{d} \in \{0, 2, 4\}$ space. After flowing through all the downsampling layers (decoder blocks), a 3×3 convolutional layer is employed to recover the spectral dimension and reconstruct the residual image $\mathbf{x}_{\mathrm{res}}$ as

$$\mathbf{x}_{\mathrm{res}} = f_{\mathrm{FRNN}}\left(\widetilde{\mathbf{F}}_{\mathcal{D}_1} \oplus \mathbf{F}_{\mathcal{D}_1}\right) \tag{15}$$

where f_{FRNN} denotes the 3×3 convolutional layer followed by BN and LeakyReLU employed at FRNN.

After carrying out DIP upsampling and residual prediction of our DIP-HyperKite, the DIP upsampled HSIs $\mathbf{x}_{\mathrm{dip}}$ and $\mathbf{x}_{\mathrm{res}}$ are created. Finally, we can obtain the fused HSI $\mathbf{x}$ by using $\mathbf{x}_{\mathrm{dip}}$ and $\mathbf{x}_{\mathrm{res}}$ as

$$\mathbf{x} = \mathbf{x}_{\mathrm{res}} + \mathbf{x}_{\mathrm{dip}}. \tag{16}$$

To this end, we utilize L_1 loss to optimize HyperKite, which has been demonstrated as a superior choice for remote sensing image SR [5], [35] and also experimentally verified to be effective for improving the fusion accuracy. For the training set $\{\mathbf{x}_{\mathrm{in}}^k, \mathbf{x}_{\mathrm{ref}}^k\}^N$, where $\mathbf{x}_{\mathrm{in}}^k$ is the kth input, $\mathbf{x}_{\mathrm{ref}}^k$ is the kth reference HSI, and N is the total number of training HSIs in the training set. The L_1 loss function utilized for HyperKite training can be defined as follows:

$$L(\Theta) = \frac{1}{N}\sum_{k=1}^{N}\left\|\left(\mathbf{x}_{\mathrm{dip}}^k + f_{\mathrm{HyperKite}}\left(\mathbf{x}_{\mathrm{in}}^k\right)\right) - \mathbf{x}_{\mathrm{ref}}^k\right\|_1. \tag{17}$$

Moreover, all the parameter details of our proposed HyperKite are summarized in Table II. We train our network in the Pytorch framework using an NVIDIA Quadro 8000 GPU. We use the Adam optimizer with a learning rate of 0.001, a weight decay of 0.0001, and a momentum of 0.9 to train HyperKite. We use a batch size of 4 and train the network for 2500 epochs.

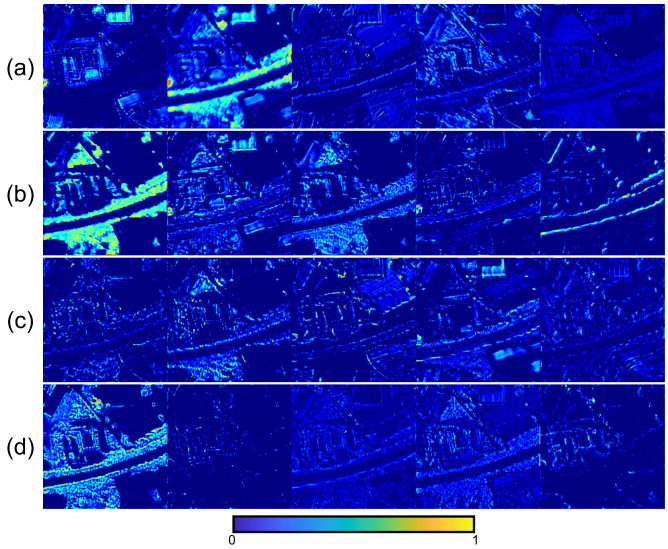


Fig. 7. Visualization of filter responses of HyperKite. (a) Feature maps from the first layer of encoder. (b) Feature maps from the second layer of encoder. (c) Feature maps from the third layer of encoder. (d) Feature maps from the third layer of encoder. By restricting the receptive field, HyperKite is able to focus on edges and smaller regions. Zoomed-in view recommended.

IV. Experimental Settings

A. Datasets

To evaluate the performance of our proposed DIP-HyperKite for HS pansharpening, we conduct a series of experiments on three semisynthetic and one real HSI datasets, which are described in detail next.

1) Pavia Center Dataset: The Pavia Center scene was captured by the ROSIS camera [55]. The original HSI consists of 115 spectral bands spanning from 430 to 960 nm. The spatial size of the original image is 1096×1096 pixels, where a single pixel is equivalent to a geometric resolution of 1.3×1.3 m^2. The 13 noisy spectral bands in the original HSI were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

discarded, thus resulting in an HSI with 102 spectral bands spanning from 430 to 860 nm. In addition, a rectangular area of size 1096 × 381 pixels with no information at the center of the original HSI was also discarded, and the resulting "two-part" image with size of 1096 × 715 × 102 was used for the experiments. Following the same experimental procedure outlined in [30], we also used only the top-left corner of the HSI with size of 960 × 640 × 102 and partitioned it into 24 cubic patches of size 160 × 160 × 102 with no overlap, which constituted the reference images ($\mathbf{x}_{\text{ref}}$) of Pavia Center dataset. In order to generate PAN images ($\mathbf{p}$) and LR-HSIs ($\mathbf{y}$) corresponding to each HR-HSI, we utilize Wald's protocol [56]. Following Wald's protocol, we generate PAN images ($\mathbf{p}$) of size 160 × 160 by averaging first 61 spectral bands of HR reference HSI. In order to generate LR-HSIs of size 40 × 40 × 102, we spatially blurred the HR reference HSI with an 8 × 8 Gaussian filter and then downsampled the result. The scaling factor (β) was set to 4 for the Pavia Center dataset. We randomly select 17 cubic patches for the training, and the rest of the seven patches forms the testing set of the Pavia Center dataset.

2) Botswana Dataset: The Botswana scene was acquired by the Hyperion sensor on the NASA's Earth Observing 1 (EO-1) satellite. The original Botswana HSI consists of 242 spectral bands spanning from 400 to 2500 nm with a spectral resolution of 10 nm. The spatial size of the original Botswana image is 1496 × 256 pixels. We remove the uncalibrated and noisy spectral bands in the original image, thus resulting in an HSI with 145 spectral bands. Following [30], we also use only the top-left corner of the HSI with size of 1200 × 240 × 145 and partitioned it into 20 cubic patches of size 120 × 120 × 145 with no overlap, which constitutes the reference images $\mathbf{x}_{\text{ref}}$ of the Botswana dataset. Next, we generate PAN images $\mathbf{p}$ of size 120 × 120 by averaging the first 31 spectral bands of HR-HSI. We utilized the same procedure mentioned for the Pavia Center dataset to generate LR-HSIs $\mathbf{y}$ except that we keep the downsampling factor β as 3. We randomly select 14 cubic patches for training, and the rest of the patches are utilized for testing.

3) Chikusei Dataset [57]: The Chikusei scene was captured by the Headwall Hyperspec-VNIR-C imaging sensor over the agricultural and urban areas in Chikusei, Japan. The original Chikusei HSI consists of 128 spectral bands spanning from 363 to 1018 nm. The spatial size of the Chikusei HSI is 2517 × 2335 pixels, where a single pixel is equivalent to a geometric resolution of 2.5 × 2.5 m^2. We used the top-left corner of the HSI with size of 2304 × 2304 × 128 and partitioned it into 81 cubic patches of size 256 × 256 × 128 with no overlap, which constituted the reference images $\mathbf{x}_{\text{ref}}$ of Chikusei dataset. Next, we generate PAN images and LR-HSIs following the same procedure mentioned for the Pavia Center dataset. We randomly select 61 cubic patches for training, and the rest of the patches are utilized for testing.

4) Los Angeles Dataset: The Los Angeles dataset is a real HSI dataset that was acquired over a port in the city of Los Angeles. This dataset consists of LR-HSI and the HR-PAN image that were captured by the Hyperion sensor on the EO-1 satellite and the Advanced Land Imager (ALI), respectively. The original LR-HSI consists of 242 spectral bands with a spatial resolution of 30 m. The spatial resolution of PAN image is 10 m. During the preprocessing, the uncalibrated and noisy spectral bands of LR-HSI were removed, thus resulting in an LR-HSI with 145 spectral bands for experimentation. Therefore, the well-trained model on the Botswana dataset can be generalized to the Los Angeles dataset. The size of the LR-HSI is 120 × 120 × 145, and the size of the experimental PAN image is 360 × 360.

Note: The standard deviation (σ) of the Gaussian filter that we use to generate LR-HSIs is calculated as [30], [58]

$$\sigma = \sqrt{\frac{1}{2 \times 2.7725887/\beta^2}} = 0.4247\beta. \tag{18}$$

B. Performance Measures

In order to evaluate the quality of the proposed pansharpening method, we use different image quality measures. Following [30], we use reference-based metrics such as cross correlation (CC), spectral angle mapping (SAM), root-mean-square error (RMSE), Errur Relative Globale Adimensionnelle Desynthese (ERGAS), and peak signal-to-noise ratio (PSNR) to evaluate the pansharpening performance on semisynthetic datasets where the reference HSI is available. For the real HSI dataset where the reference image is not available to evaluate the above performance measures, we adopt no-reference-based performance metrics, such as spectral distortion (D_λ), spatial distortion (D_S), and quality with no-reference (QNR). These measures have been widely used in the HSI processing community and are appropriate for evaluating fusion in spectral and spatial resolutions.

1) CC: The CC metric characterizes the geometric distortion and is defined as

$$\text{CC}(\mathbf{x}, \mathbf{x}_{\text{ref}}) = \frac{1}{l}\sum_{i=1}^{l} \text{CCS}\left(\mathbf{x}^i, \mathbf{x}_{\text{ref}}^i\right) \tag{19}$$

where CCS denotes the cross correlation for a single-band image as follows:

$$\text{CCS}(\mathbf{A}, \mathbf{B}) = \frac{\sum_{j=1}^{n}\left(\mathbf{A}_j - \mu_A\right)\left(\mathbf{B}_j - \mu_B\right)}{\sqrt{\sum_{j=1}^{n}\left(\mathbf{A}_j - \mu_A\right)^2 \sum_{j=1}^{n}\left(\mathbf{B}_j - \mu_B\right)^2}} \tag{20}$$

where n is the total number of pixels in the image and $\mu_A = (1/n)\sum_{j=1}^{n}\mathbf{A}_j$ is the sample mean of $\mathbf{A}$. The ideal value of CC is 1.0, which indicates that the two HSIs are highly correlated.

2) SAM: SAM is a spectral measure, which is defined as

$$\text{SAM}(\mathbf{x}, \mathbf{x}_{\text{ref}}) = \frac{1}{n}\sum_{j=1}^{n} \text{SAM}\left(\mathbf{x}_j, \mathbf{x}_{\text{ref}\,j}\right) \tag{21}$$

where given the vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^l$

$$\text{SAM}(\mathbf{a}, \mathbf{b}) = \arccos\left(\frac{\langle \mathbf{a}, \mathbf{b}\rangle}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}\right) \tag{22}$$

where $\langle \mathbf{a}, \mathbf{b}\rangle$ denotes the inner product between $\mathbf{a}$ and $\mathbf{b}$ and $\|\cdot\|_2$ is the L_2 norm. The SAM is a measure of spectral shape

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

preservation. The SAM values reported in our experiments are in degrees and thus belong to $(-90, 90]$. The optimal value of SAM is 0.0. The values of SAM reported in our experiments have been obtained by averaging the values for all image pixels.

3) RSNR/RMSE: The reconstruction SNR (RSNR) or RMSE is related to the difference between the reference and fuse images, which is defined as follows:

$$\mathrm{RMSE}(\mathbf{x}, \mathbf{x}_{\mathrm{ref}}) = \frac{1}{n \times l}\|\mathbf{x} - \mathbf{x}_{\mathrm{ref}}\|_F^2 \tag{23}$$

$$\mathrm{RSNR}(\mathbf{x}, \mathbf{x}_{\mathrm{ref}}) = 10\log_{10}\left(\frac{\|\mathbf{x}_{\mathrm{ref}}\|_F^2}{\|\mathbf{x} - \mathbf{x}_{\mathrm{ref}}\|_F^2}\right). \tag{24}$$

4) ERGAS: Relative dimensionless global error in synthesis (ERGAS) calculates the amount of spectral distortion in the image. The ERGAS measure is defined as

$$\mathrm{ERGAS} = 100\frac{1}{d^2}\sqrt{\frac{1}{l}\sum\nolimits_{i=1}^{l}\left(\frac{\mathrm{RMSE}\left(\mathbf{x}^i, \mathbf{x}_{\mathrm{ref}}^i\right)}{\mu\left(\mathbf{x}_{\mathrm{ref}}^i\right)}\right)} \tag{25}$$

where d is the ratio between the linear resolution of the PAN image and the HSIs defined as

$$d = \frac{\text{PAN linear spatial resolution}}{\text{HS linear spatial resolution}} \tag{26}$$

where $\mathrm{RMSE}(\mathbf{x}^i, \mathbf{x}_{\mathrm{ref}}^i) = ((\|\mathbf{x}^i - \mathbf{x}_{\mathrm{ref}}^i\|_F)/\sqrt{n})$ and $\mu(\mathbf{x}_{\mathrm{ref}}^i)$ is the sample mean of the ith band of $\mathbf{x}_{\mathrm{ref}}$. The ideal value of ERGAS is 0.

5) PSNR: The PSNR also assesses the fusion quality of each spectral bands, and the average PSNR is calculated as

$$\mathrm{PSNR} = \frac{1}{l}\sum_{i=1}^{l}\left[10\log_{10}\left(\frac{\max\left(\mathbf{x}_{\mathrm{ref}}^i\right)}{\mathrm{RMSE}\left(\mathbf{x}^i, \mathbf{x}_{\mathrm{ref}}^i\right)}\right)^2\right] \tag{27}$$

where $\max(\mathbf{x}_{\mathrm{ref}}^i)$ is the maximum pixel value in the ith band of $\mathbf{x}_{\mathrm{ref}}$. A larger value of PSNR indicates a higher reconstruction quality in spatial information of the fusion result.

6) QNR: To quantitatively evaluate the pansharpening performance on real datasets that do not have reference HSI, we adopt the QNR metric [30]. The QNR metric can be defined as

$$QNR = (1 - D_\lambda)^\eta (1 - D_S)^\rho \tag{28}$$

where D_λ defines the amount of spectral distortion, D_S quantifies the amount of spatial distortion, and η and ρ are two coefficients (usually set to 1). The amount of spectral distortion D_λ is calculated as

$$D_\lambda = \sqrt[\epsilon]{\frac{1}{l(l-1)}\sum\nolimits_{q_1=1}^{l}\sum\nolimits_{q_2=1, q_2\neq q_1}^{l}\left\|Q_{\mathbf{y}}^{q_1,q_2} - Q_{\mathbf{x}}^{q_1,q_2}\right\|^\epsilon} \tag{29}$$

where parameter ϵ is usually set to 1. $Q_{\mathbf{y}}^{q_1,q_2} = Q(\mathbf{y}^{q_1}, \mathbf{y}^{q_2})$ and $Q_{\mathbf{x}}^{q_1,q_2} = Q(\mathbf{x}^{q_1}, \mathbf{x}^{q_2})$ are denoted as the Q-*index* [59], which calculates the dissimilarities between couples of spectral bands for LR-HSI $\mathbf{y}$ and pansharpen HSI $\mathbf{x}$. The spatial distortion D_S is calculated as

$$D_S = \sqrt[\delta]{\frac{1}{l}\sum\nolimits_{q=1}^{b}\|Q(\mathbf{x}^q, \mathbf{p}) - Q(\mathbf{y}^q, \mathbf{p}_{\mathrm{lr}})\|^\delta} \tag{30}$$

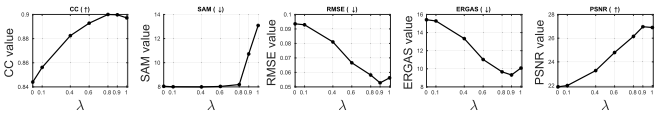


Fig. 8. Variation of CC, SAM, RMSE, ERGAS, and PSNR with the regularization constant λ in our spectral + spectral energy function Q_{ss} for the Pavia Center dataset. We select $\lambda = 0.8$ as the optimal value of regularization constant for the Pavia Center dataset by considering all the performance metrics.

where δ is typically set to 1 and $\mathbf{p}_{\mathrm{lr}}$ denotes the simulated LR-PAN image with the same size of LR-HSI. The ideal values of D_λ, D_S, and QNR are 0, 0, and 1, respectively.

V. Results and Discussion

This section presents the results of our proposed DIP-HyperKite for HS pansharpening and compares it with the state-of-the-art methods on the Pavia Center, Botswana, and Chikusei datasets. For better clarity, we divide this section into two parts. In the first part (Section V-A), we highlight the contribution from our proposed spatial + spectral energy function for the DIP upsampling process and compare it with available state-of-the-art upsampling techniques such as nearest neighbor, bicubic, LapSRN, and DIP with only spectral loss. In the second part (Section V-B), we present the final fusion results that we obtain from our proposed HyperKite network and compare it with classical and deep learning-based pansharpening approaches.

A. Effect of the Proposed Spatial + Spectral Energy Function for the DIP Upsampling Process

As we discussed in Section III-A, the recently proposed pansharpening methods, such as DHP-DARN [30] and DHP [33], utilized the DIP process to upsample the LR-HSI instead of using the nearest-neighbor, bicubic, or LapSRN techniques due to its excellent performance. However, we have observed that the quality of upsampled HSI can be further improved by carefully redesigning the loss function used in the DIP optimization. Instead of only utilizing spectral constraint in the DIP loss function, we derived a novel loss function with spectral and spatial constraints. This section demonstrates the performance improvement brought by our proposed spatial + spectral loss function to the DIP upsampling process. We compare DIP with the proposed spatial + spectral loss against the DIP with spectral loss only. Furthermore, to make the analysis more comprehensive, we also added conventional upsampling techniques used in the HS pansharpening domain, such as nearest neighbor and bicubic. Furthermore, motivated by the experimental discussion in [30], we also added the results from LapSRN [32], which is trained on a large amount of RGB images.

1) Tuning the Hyperparameter λ in Our Spatial + Spectral Energy Function: We start our discussion with the effect of the regularization constant λ in our proposed spatial + spectral loss function as defined in (5). The variation of CC, SAM, RMSE, ERGAS, and PSNR values when varying the regularization parameter λ from 0.0 to 1.0 for the Pavia Center, Botswana, and Chikusei datasets is shown in Figs. 8–10,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

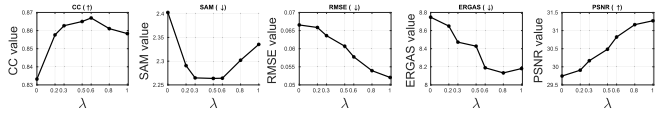


Fig. 9. Variation of CC, SAM, RMSE, ERGAS, and PSNR with the regularization constant λ in our spectral + spectral energy function Q_{ss} for the Botswana dataset. We select $\lambda = 0.8$ as the optimal value of regularization constant for the Botswana dataset by considering all the performance metrics.

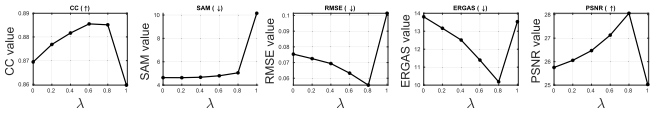


Fig. 10. Variation of CC, SAM, RMSE, ERGAS, and PSNR with the regularization constant λ in our spectral + spectral energy function Q_{ss} for the Chikusei dataset. We select $\lambda = 0.8$ as the optimal value of regularization constant for the Chikusei dataset by considering all the performance metrics.

respectively. As can be seen from these figures, as the value of the regularization constant λ increases, the performance metrics also begin to improve, then hit a saturation point, and then degrade for all three datasets. Therefore, we carefully select the regularization constant λ for each dataset by considering all the performance metrics. For example, consider the variation of the performance metrics with the regularization constant λ for the Pavia Center dataset, which is shown in Fig. 8. As we can see, when the value of the regularization constant increases from 0.0 to 0.8, we can see that CC, RMSE, ERGAS, and PSNR start to improve, and when λ increases beyond 0.8, the performance metrics start to degrade. Therefore, we set $\lambda = 0.8$ as the optimal value of the regularization constant of our proposed spatial + spectral energy term for the Pavia Center dataset. The variation in performance metrics with the regularization parameter λ for the Botswana and Chikusei datasets is also shown in Figs. 9 and 10, respectively. Following the same analysis we described for the Pavia Center dataset, we select $\lambda = 0.8$ as the optimal value of the regularization constant for the Botswana and Chikusei datasets. Note that the performance improvement brings from our proposed spatial + spectral loss function for the DIP upsampling process. Under the optimal regularization constant ($\lambda = 0.8$), our spatial + spectral energy function improves the quality of upsampled HSIs over the spectral loss (equivalent to $\lambda = 0$ point in Figs. 8–10) in terms of CC, RMSE, ERGAS, and PSNR metrics by 6.64%, 37.8%, 63.4%, 37.3%, and 19.5%, respectively, for the Pavia Center dataset. For the Botswana dataset, our proposed loss function improves CC, SAM, RMSE, ERGAS, and PSNR metrics over the DIP with spectral loss by 3.3%, 4.2%, 19.1%, 14.7%, 7.0%, and 5.2%, respectively. Similarly, for the Chikusei dataset, our method improves CC, RMSE, ERGAS, and PSNR metrics compared to DIP with spectral loss by 1.8%, 26.3%, 22.9%, 26.2%, and 8.9%, respectively.

Discussion on the Regularization Constant λ: Let us first consider the case where the regularization constant λ is set to zero. This is equivalent to the case where we only have the spectral constraint. In this case, the DIP network minimizes the distance between the downsampled version of the upsampled HSI and the LR-HSI. Since the downsampling operator acts as a low-pass filter in the frequency domain, what DIP network actually minimizes is that the distance between the low-pass version of the upsampled HSI and the LR-HSI. Because of this reason, the upsampled HSI from the DIP network trained only with spectral constraint lacks the high-frequency components, such as edge information and fine structures. Now, let us consider the case where we have both spatial and spectral constraints in the DIP loss function. As we described in Section III-A, we combined the spatial and spectral constraints via regularization parameter λ. The value of λ controls the fidelity of the predicted PAN image toward the actual PAN image. Since the predicted PAN image and the upsampled HSI are coupled via SRF, to make the predicted PAN image close as possible to the actual PAN image, the DIP network tries to predict some of the high-frequency components such as edges and fine structures in the PAN image while maintaining the low-pass version of the upsampled HSI close to the LR-HSI. Therefore, the regularization constant what actually controls is the amount of high-frequency components fused from the PAN image to the upsampled HSI. This explains the observation that we made from Figs. 8 to 10, where when the value of the regularization parameter increases, the DIP network embeds some of the high-frequency information to the upsampled HSI, which ultimately helps to improve the quality of the upsampled image. However, when the value of the regularization constant is large, the spatial loss term starts to dominate the loss function, resulting in drop of spectral-domain performance metrics such as SAM and ERGAS. Therefore, we can achieve high-quality upsampled HSIs by appropriately controlling the regularization parameter in the spatial + spectral energy function.

TABLE III

AVERAGE QUANTITATIVE RESULTS FOR DIFFERENT UPSAMPLING TECHNIQUES ON THE PAVIA CENTER DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-1}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
Nearest-neighbor	0.809	7.70	1.22	9.63	19.97	19.65
Bicubic	0.840	7.45	1.13	11.26	18.48	20.36
LapSRN [32]	0.843	**7.37**	1.12	11.56	18.16	20.49
DIP+spectral [30]	0.844	8.04	0.94	14.91	15.42	21.89
DIP+Q_{ss} (ours)	**0.900**	8.18	**0.58**	**24.36**	**9.66**	**26.15**
	(+6.6%)	-	(-37.8%)	(+63.4%)	(-37.3%)	(+19.5%)

2) Comparison of DIP With the Proposed Spatial + Spectral Loss With State-of-the-Art Upsampling Techniques: In the previous section, we determined the optimal value of the regularization constant λ for our proposed spatial + spectral loss function for the three datasets. In this section, we compare DIP with our spatial + spectral loss against DIP with only spectral loss and other commonly used upsampling techniques such as nearest neighbor, bicubic, and LapSRN, both qualitatively and quantitatively.

Table III summarizes the quantitative results of nearest neighbor, bicubic, LapSRN, and DIP upsampling methods for the Pavia Center dataset. For this dataset, our proposed DIP method improves the quality of upsampled images in terms of CC, SAM, RSNR, ERGAS, and PSNR performance measures by 6.6%, 37.3%, 63.4% 37.3%, and 19.5%, respectively. We have also noticed that this improvement is accompanied by a drop in the SAM index, which is around 1.8% compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

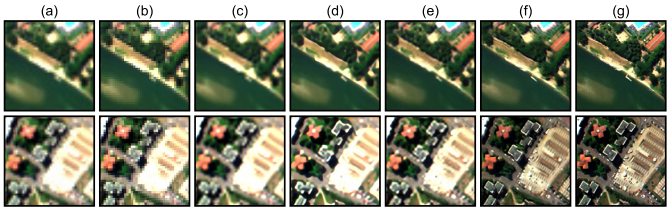


Fig. 11. Upsampled images of the 1st patch (in the first row) and 11th patch (in the second row) of Pavia Center dataset. (a) LR-HSI. (b) Nearest neighbor. (c) Bicubic. (d) LapSRN [32]. (e) DIP with only spectral energy [30]. (f) DIP with our spatial + spectral energy (Q_{ss}; $\lambda = 0.8$). (g) Reference. The RGB image is generated by utilizing the 10th, 30th, and 60th bands of the HSI for blue, green, and red bands, respectively.

TABLE IV

AVERAGE QUANTITATIVE RESULTS FOR DIFFERENT UPSAMPLING TECHNIQUES ON THE BOTSWANA DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-2}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
Nearest-neighbor	0.854	2.52	7.87	29.03	9.08	28.87
Bicubic	0.852	2.42	7.67	29.60	8.77	29.17
LapSRN [32]	0.858	2.47	6.27	34.01	8.27	29.01
DIP+spectral [30]	0.833	2.40	6.66	32.91	8.75	29.75
DIP+Q_{ss} (ours)	**0.861**	**2.30**	**5.39**	**37.80**	**8.13**	**31.28**
	(+0.4%)	(-4.2%)	(-14.0%)	(+11.2%)	(-1.7%)	(+5.2%)

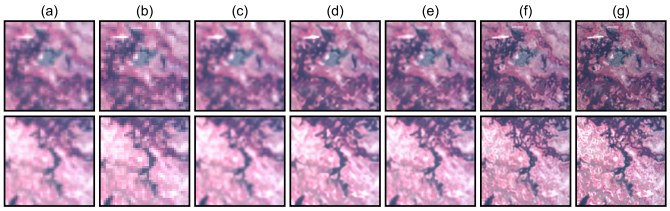


Fig. 12. Upsampled images of the 12th (in the first row) and 14th (in the second row) patch of Botswana dataset. (a) LR-HSI. (b) Nearest neighbor. (c) Bicubic. (d) LapSRN [32]. (e) DIP with only spectral energy [30]. (f) DIP with our spatial + spectral energy (Q_{ss}; $\lambda = 0.8$). (g) Reference. The RGB image is generated by utilizing the 10th, 35th, and 61th bands of the HSI for blue, green, and red bands, respectively.

to the DIP with spectral loss. This fall in the SAM index is not that significant compared to the improvements we have achieved in terms of all other performance measures. Furthermore, we can cross-verify these quantitative results with qualitative results that we have shown in Fig. 11 for the Pavia Center dataset. We can see that the DIP upsampled images with our proposed spatial + spectral constraint look much more closer to the reference image and has predicted very fine structures and edges compared to other upsampling methods.

We also summarize the quantitative results for different upsampling methods for the Botswana dataset in Table IV. As we can see, DIP with the proposed spatial + spectral loss improves the quality of upsampled images in terms of all the performance metrics by a significant margin: the CC value increased by 0.4%, the SAM value reduced by 4.3%, the RMSE value reduced by 14.0%, the RSNR value improved by 11.2%, the ERGAS value reduced by 1.7%, and the PSNR value increased by 5.2%. Also, we can verify these quantitative results with the qualitative results shown in Fig. 12 for the Botswana dataset. Similar to the qualitative results that we have observed for the Pavia Center dataset, we can see that the upsampled images using DIP with our proposed spatial + spectral loss are much more closer to the reference HSI.

TABLE V

AVERAGE QUANTITATIVE RESULTS FOR DIFFERENT UPSAMPLING TECHNIQUES ON THE CHIKUSEI DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-2}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
Nearest-neighbor	0.861	4.05	9.99	18.26	17.03	23.73
Bicubic	0.884	3.86	9.31	20.07	15.75	24.52
LapSRN [32]	0.885	**3.75**	8.53	21.37	14.33	25.06
DIP+spectral [30]	0.869	4.64	7.54	24.16	13.80	25.75
DIP+Q_{ss}(ours)	**0.885**	5.05	**5.56**	**29.69**	**10.18**	**28.06**
	(+0.1%)	-	(-26.3%)	(+22.9%)	(-26.2%)	(+8.9%)

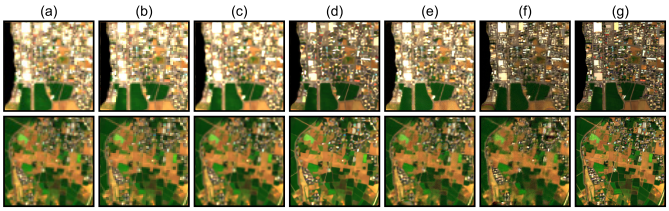


Fig. 13. Upsampled images of the 37th (in the first row) and 50th (in the second row) patch of Chikusei dataset. (a) LR-HSI. (b) Nearest neighbor. (c) Bicubic. (d) LapSRN [32]. (e) DIP with only spectral energy [30]. (f) DIP with our spatial + spectral energy (Q_{ss} ($\lambda = 0.8$). (g) Reference. The RGB image is generated by utilizing the 12th, 20th, and 29th bands for blue, green, and red bands, respectively.

Finally, we summarize the quantitative results for different upsampling methods for the Chikusei dataset in Table V. In this case also, the performance of DIP upsampled images with our proposed spatial + spectral loss outperforms five out of six performance measures that we considered for the analysis. As we can see from Table V, our DIP method has increased the value of CC by 0.1%, has decreased the value of RMSE by 26.3%, has increased the RSNR by 22.9%, has decreased the ERGAS by 26.2%, and has increased the PSNR by 8.9% over the state-of-the-art results. Similar to the Pavia Center dataset, in this dataset, also we have observed the drop in SAM index; however, this is negligible compared to the performance gained in terms of the other quantitative measures. Furthermore, following the similar trend with other datasets, we have included the qualitative results in Fig. 13 for the Chikusei dataset. From the qualitative results, also, we can see that DIP with our spatial + spectral constraint is able to predict very fine structures and edges more accurately than the other methods.

In summary, we have shown that the DIP method with our proposed spatial + spectral constraints outperforms the state-of-the-art upsampling methods with a significant margin in all the datasets that we have considered in this study. In the next section, we present the final fusion results and compare them with state-of-the-art pansharpening algorithms, qualitatively and quantitatively.

B. Final Fusion Results on Semisynthetic HS Datasets

In this section, we compare the final fusion results from our DIP-HyperKite with the state-of-the-art pansharpening approaches, such as PCA [13], GFPCA [21], BF [23],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
AVERAGE QUANTITATIVE RESULTS ON THE PAVIA CENTER DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-2}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
PCA [13]	0.845	8.92	3.45	34.32	6.64	31.26
GFPCA [21]	0.902	8.31	3.98	29.34	7.44	29.09
BF [23]	0.918	9.60	3.44	31.99	6.63	30.22
BFS [24]	0.925	8.10	3.05	34.37	6.00	31.09
SFIM [18]	0.946	6.76	2.55	37.47	5.43	32.61
GS [11]	0.961	6.62	2.55	38.08	4.95	32.93
GSA [11]	0.950	7.15	2.34	39.60	4.70	33.52
MTF-GLP-HPM [20]	0.955	6.81	2.25	40.70	4.77	33.97
CNMF [25]	0.960	6.64	2.20	40.79	4.39	34.14
MTF-GLP [19]	0.956	6.55	2.20	40.70	4.45	34.12
HySure [22]	0.966	6.13	1.80	44.60	3.77	35.91
HyperPNN [35]	0.967	6.09	1.67	48.62	3.82	36.70
DHP-DARN [30]	0.969	6.43	1.56	49.17	3.95	37.30
DIP-HyperKite (ours)	**0.980**	**5.61**	**1.29**	**51.72**	**2.85**	**38.65**

TABLE VII
AVERAGE QUANTITATIVE RESULTS ON THE BOTSWANA DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-2}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
PCA [13]	0.946	2.22	1.74	57.10	2.89	28.17
GFPCA [21]	0.925	2.48	1.97	53.81	3.18	26.75
BF [23]	0.919	2.41	1.86	55.43	3.37	26.88
BFS [24]	0.918	2.39	1.85	55.52	3.38	26.91
SFIM [18]	0.890	3.31	2.56	48.30	2.98	27.27
GS [11]	0.949	2.17	1.68	57.55	2.74	28.32
GSA [11]	0.964	1.86	1.28	63.02	2.16	30.78
MGH [20]	0.962	1.90	1.33	62.23	2.15	30.47
CNMF [25]	0.951	2.28	1.38	60.90	2.48	29.63
MG [19]	0.963	1.88	1.32	62.23	2.16	30.45
HySure [22]	0.963	1.93	1.19	63.80	2.12	30.97
HyperPNN [35]	0.957	1.92	1.06	66.22	2.40	29.00
DHP-DARN [30]	0.954	1.91	1.05	66.22	2.35	29.98
DIP-HyperKite (ours)	**0.974**	**1.68**	**0.96**	**67.98**	**1.89**	**32.12**

BFS [24], SFIM [18], GS [11], GSA [11], MTF-GLP-HPM [20], CNMF [25], MTF-GLP [19], HySure [22], HyperPNN [35], and DHP-DARN [30] for Pavia Center, Botswana, and Chikusei datasets.

1) Final Fusion Results on the Pavia Center Dataset: The average quantitative results for different pansharpening approaches on the testing set of the Pavia Center dataset are shown in Table VI. As can be seen from Table VI, our proposed HyperKite achieves the highest CC value compared to all the other pansharpening approaches that we have considered in this study. A higher CC value indicates that the fused HSI is closer to the actual HSI with less geometric distortion. Furthermore, our proposed DIP-HyperKite achieved the smallest values for SAM, RMSE, and ERGAS performance measures, indicating the best fusion performance over the other pansharpening approaches. Especially, the smallest SAM and ERGAS indicate that our DIP-HyperKite can fuse HSIs with less spectral distortion than the state-of-the-art methods. In addition, our DIP-HyperKite improved the PSNR metric by 3.6% over the state-of-the-art value. To further verify the fusion quality of our proposed DIP-HyperKite, we present qualitative results in Fig. 14 for the Pavia Center dataset. To better highlight the fusion quality between different pansharpening approaches, we have shown the mean absolute error (MAE) plots along with the RGB composite image for

TABLE VIII
AVERAGE QUANTITATIVE RESULTS ON THE CHIKUSEI DATASET

Method	CC (↑)	SAM (↓)	RMSE $\times 10^{-2}$ (↓)	RSNR (↑)	ERGAS (↓)	PSNR (↑)
GFPCA [21]	0.883	4.76	1.98	34.22	7.00	37.05
BF [23]	0.903	5.15	1.94	34.40	6.62	37.89
BFS [24]	0.917	4.69	1.72	36.84	6.39	37.99
SFIM [18]	0.928	3.79	1.43	40.51	6.43	39.55
GS [11]	0.733	5.64	2.96	26.37	8.17	35.13
GSA [11]	0.943	3.52	1.42	40.73	4.30	41.38
MTF-GLP-HPM [20]	0.929	3.82	1.45	39.38	6.40	39.85
CNMF [25]	0.900	4.72	1.91	36.73	5.75	39.65
MTF-GLP [19]	0.938	3.81	1.52	39.38	4.41	41.05
HySure [22]	0.960	2.98	1.13	45.24	3.69	43.14
HyperPNN [35]	0.946	3.97	1.11	46.55	4.77	41.57
DHP-DARN [30]	0.953	3.60	1.05	46.66	4.44	42.24
DIP-HyperKite (ours)	**0.974**	**2.85**	**1.03**	**46.97**	**3.62**	**43.53**

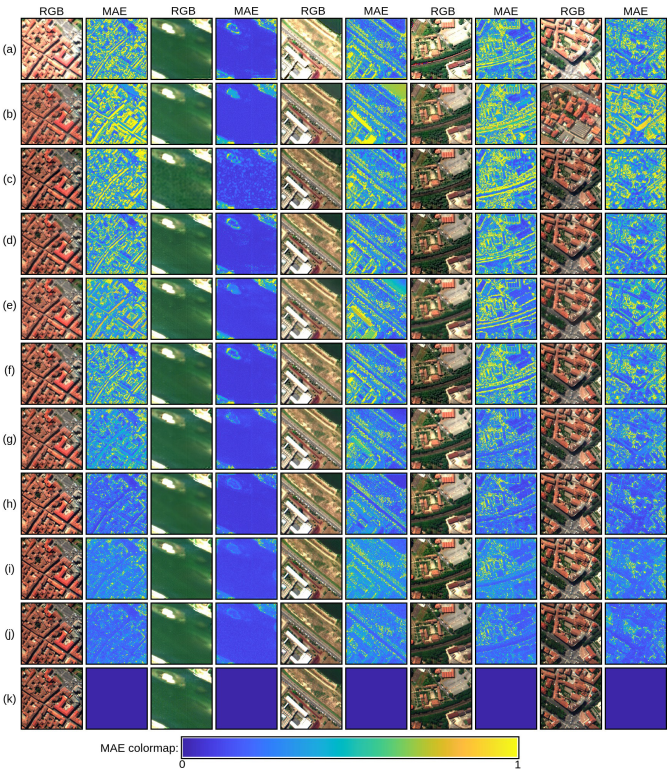


Fig. 14. Visual results generated by different pansharpening algorithms for the 1st (in the first and second columns), 3rd (in the third and fourth columns), 12th (in the fifth and sixth columns), 20th (in the seventh and eight columns), and 21st (in the nine and tenth columns) patches of the Pavia Center dataset. (a) SFIM [18]. (b) GS [11]. (c) GSA [11]. (d) MTF-GLP-HPM [20]. (e) CNMF [25]. (f) MTF-GLP [19]. (g) HySure [22]. (h) HyperPNN [35]. (i) DHP-DARN [30]. (j) DIP-HyperKite (ours). (k) Reference.

each fused HSI. According to the figure, the MAE maps corresponding to our DIP-HyperKite are much purple than the other pansharpening approaches, indicating minor fusion error. This is mainly because of the ability of our HyperKite network to predict very fine structures and edges by constraining the receptive field of the deep network.

2) Final Fusion Results on the Botswana Dataset: Table VII summarizes the average quantitative results of different fusion methods on the Botswana dataset. Similar to the Pavia Center dataset, we can see that our DIP-HyperKite outperforms all the other HS pansharpening approaches by a considerable margin.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

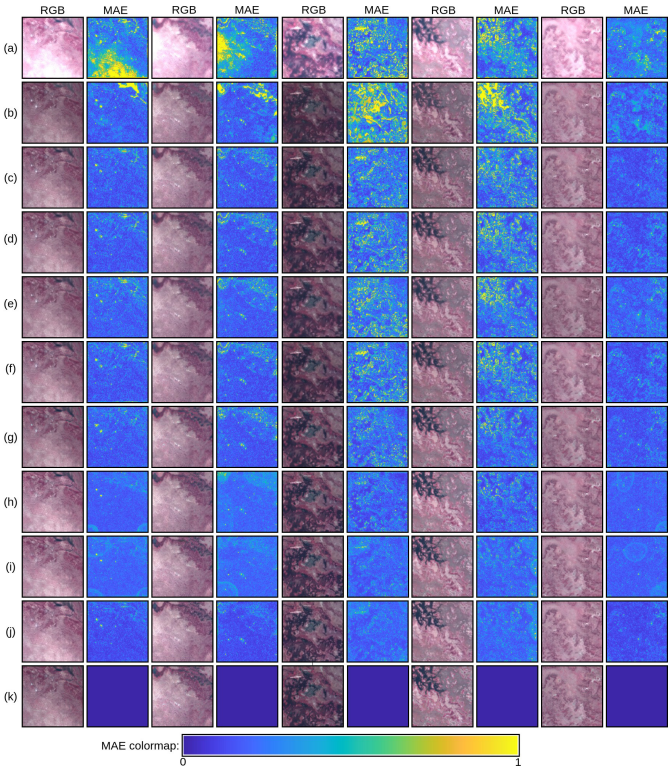


Fig. 15. Visual results generated by different pansharpening algorithms for the 1st (in the first and second columns), 4th (in the third and fourth columns), 12th (in the fifth and sixth columns), 16th (in the seventh and eight columns), and 19th (in the nine and tenth columns) patches of the Botswana dataset. (a) SFIM [18]. (b) GS [11]. (c) GSA [11]. (d) MTF-GLP-HPM [20]. (e) CNMF [25]. (f) MTF-GLP [19]. (g) HySure [22]. (h) HyperPNN [35]. (i) DHP-DARN [30]. (j) DIP-HyperKite (ours). (k) Reference.

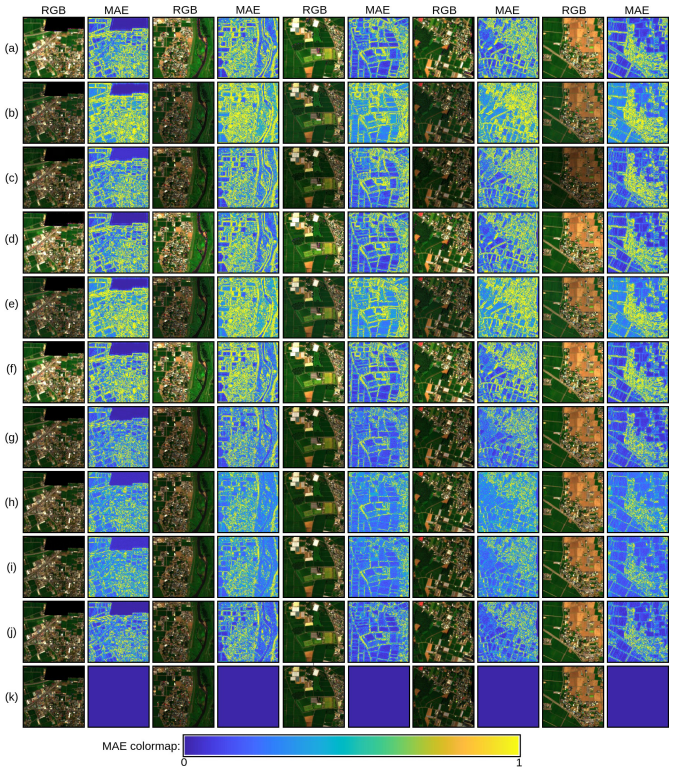


Fig. 16. Visual results generated by different pansharpening algorithms for the 5th (in the first and second columns), 13th (in the third and fourth columns), 16th (in the fifth and sixth columns), 27th (in the seventh and eight columns), and 32nd (in the nine and tenth columns) patches of the Chikusei dataset. (a) SFIM [18]. (b) BF [23]. (c) GSA [11]. (d) MTF-GLP-HPM [20]. (e) BFS [24]. (f) MTF-GLP [19]. (g) HySure [22]. (h) HyperPNN [35]. (i) DHP-DARN [30]. (j) DIP-HyperKite (ours). (k) Reference.

Concretely, our DIP-HyperKite has improved the CC by 1.03% and PSNR by 3.71%. In addition, our method has reduced the SAM by 9.68%, RMSE by 8.57%, and ERGAS by 10.85%. Furthermore, we have shown the qualitative results related to different pansharpening approaches on the Botswana dataset in Fig. 15. By observing the RGB images and MAE plots in Fig. 15, we can see that the fusion results related to our method are much closer to the reference image than the other pansharpening approaches.

3) Final Fusion Results on the Chikusei Dataset: In this section, we compare the qualitative and quantitative results on the Chikusei dataset. The average quantitative results of different pansharpening approaches on the Chikusei dataset are listed in Table VIII. Similar to the results we have observed for the other two datasets, for this dataset, also our proposed DIP-HyperKite outperforms all the pansharpening approaches that we considered for the analysis. Our pansharpening method improves the CC, SAM, RMSE, RSNR, ERGAS, and PSNR performance measures over the state-of-the-art results by 1.45%, 19.0%, 6.67%, 0.67%, 18.5%, and 0.90%, respectively. To further highlight the fusion quality of our method, we present the qualitative results of selected panshaprpening approaches for the Chikusei dataset shown in Fig. 16. By observing the RGB composite image and the MAE maps, we can clearly see that the fusion quality of the proposed DIP-HyperKite is higher than those of the other pansharpening approaches.

Variation of MAE With Each Spectral Band: We visualize the variation of MAE with the spectral band for a randomly selected test image from the testing set of Pavia Center, Botswana, and Chikusei datasets in Fig. 17 to further highlight the difference between the proposed DIP-HyperKite and state-of-the-art pansharpening method—DHP-DARN [30]. As shown in Fig. 17, the proposed DIP-HyperKite results in overall low MAE across the spectral bands compared to the DHP-DARN. This further demonstrates the low spectral and spatial distortions introduced by our proposed DIP-HyperKite.

C. Final Fusion Results on a Real Hyperspectral Dataset

This section demonstrates the generalization capability of our proposed HyperKite on a real HSI dataset: the Los Angeles dataset. Since the Los Angeles dataset does not have reference HSI to compute the reference-based quantitative measures, we utilize three no-reference-based metrics following the previous works [30], namely, D_λ, D_S, and QNR as described in Section IV-B.

To obtain the pansharpen results on the Los Angeles dataset, we utilize the DIP-HyperKite trained on the Botswana dataset. The no-reference quantitative results and qualitative results for the Los Angeles dataset are presented in Table IX and Fig. 18, respectively. The two Bayesian approaches that we previously used for comparisons on simulated HSI datasets are excluded because they require a corresponding blur matrix

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

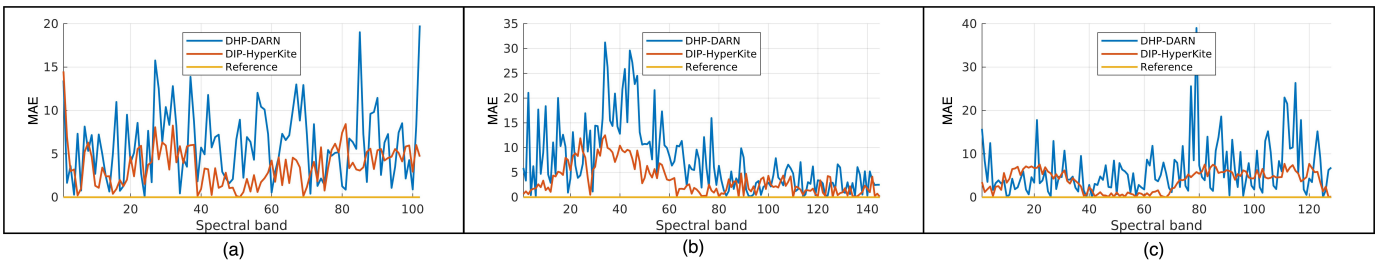


Fig. 17. Variation of MAE with spectral band for the 3rd, 8th, and 32nd patch of (a) Pavia Center, (b) Botswana, and (c) Chikusei dataset, respectively.

TABLE IX
QUANTITATIVE RESULTS (NO-REFERENCE METRICS) FOR DIFFERENT PANSHARPENING METHODS ON THE LOSS ANGELES DATASET

Metric	GSA	PCA	MG	MGH	GFPCA	CNMF	HySure	HyperPNN	DHP-DARN	DIP-HyperKite
$D_\lambda(\to 0)$	0.0750	0.0873	0.0478	0.0481	0.0697	0.0930	0.0529	0.0580	0.0459	**0.0431**
$D_S(\to 0)$	0.0942	0.1326	0.0432	0.0700	0.1891	0.1692	0.0485	0.0391	0.0372	**0.0347**
$QNR(\to 1)$	0.8378	0.7916	0.9111	0.8853	0.7544	0.7535	0.9011	0.9052	0.9186	**0.9240**

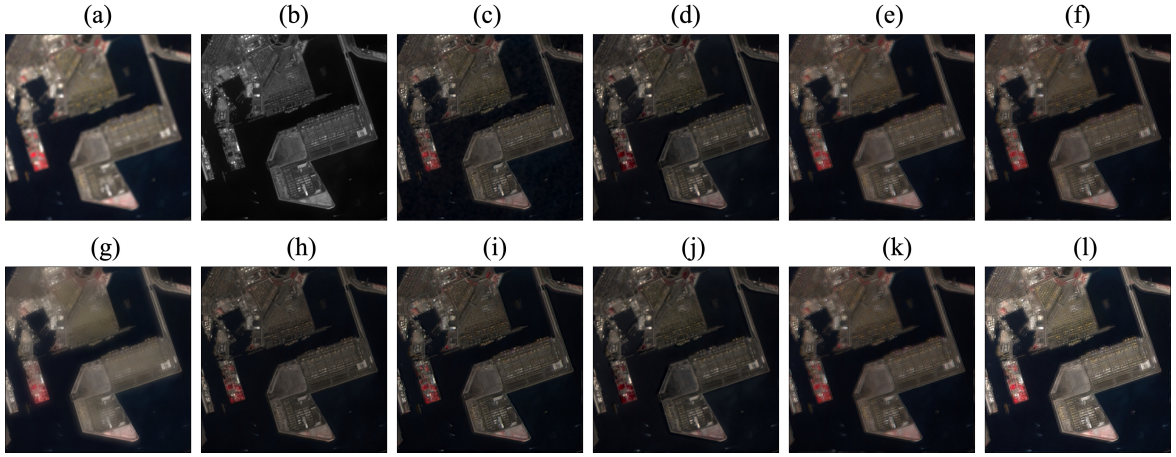


Fig. 18. Visual results obtained by different pansharpening methods for the Los Angeles dataset. (a) Real LR-HSI. (b) Real PAN image. (c) GSA [10]. (d) PCA [13]. (e) MG [19]. (f) MGH [20]. (g) GFPCA [21]. (h) CNMF [25]. (i) HySure [22]. (j) HyperPNN [35]. (k) DHP-DARN [30]. (l) DIP-HyperKite (ours). Note that the LR-HSI is zoomed-in view for visualization.

TABLE X
AVERAGE INFERENCE TIME PER HSI FOR DIFFERENT PANSHARPENING ALGORITHMS ON TESTING SET OF BOTSWANA DATASET

Method	PCA	GFPCA	BF	BFS	SFIM	GS	GSA	MGH	CNMF	MG	HySure	Hyper-PNN	DHP-DARN	DIP	Hyper-Kite	DIP-HyperKite
Time(ms)	216.8	671.9	90.2	15536.1	222.6	191.6	227.0	347.2	437.5	341.6	3541.5	0.8	58305.2	58302.9	1.5	58304.4

as an input to the algorithm, which is not available for the full-scale validation of the Los Angeles dataset.

According to the no-reference-based quantitative results shown in Table IX, we can see that our proposed DIP-HyperKite achieves outstanding results compared to the existing pansharpening approaches. These quantitative results can be further validated with the qualitative results shown in Fig. 18. All these qualitative and visual results demonstrate the high potential and generalization capability of our proposed DIP-HyperKite for HS pansharpening.

D. Inference Time

Table X presents the average inference time per HSI on the testing set of Botswana dataset for different pansharpening methods. As we discussed previously, the proposed DIP-HyperKite consists of two steps where we first upsample LR-HSI via DIP process and then predict the residual image via HyperKite. Therefore, in Table X, we present the average inference time for each step separately (i.e., DIP and HyperKite). As can be seen from Table X, we observe relatively high inference time for our DIP-HyperKite and DHP-DARN [30] methods because they both utilize DIP for upsampling, which needs to be optimized for each LR-HSI separately during the testing. In average, the DIP upsampling process takes about 58 s (58 000 ms) to upsample an LR-HSI with spatial size of 40×40 by a scaling factor of 3 as shown in Table X for the Botswana dataset.

VI. CONCLUSION

In this article, we have presented a novel approach for HS pansharpening, which mainly consists of three steps: 1) upsampling the LR-HSI via DIP; 2) predicting the residual image via overcomplete HyperKite; and 3) obtaining the final fused HSI by summation. The previously proposed DIP methods for HS upsampling only impose a constraint in the spectral domain by utilizing LR-HSI. To better preserve both spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

and spectral information, we first exploited an additional spatial constraint to DIP by utilizing the available PAN image, thereby introducing both spatial and spectral constraints to the DIP. The comprehensive experiments conducted on three HS datasets showed that our proposed spatial + spectral loss function significantly improved the quality of upsampled HSIs in CC, RMSE, RSNR, SAM, ERGAS, and PSNR performance measures. Next, in the residual prediction task, we have shown that the residual component between upsampled HSI and the reference HSI primarily consists of edge information and very fine structures. Motivated by this observation, we proposed a novel overcomplete deep learning network for the residual prediction task. In contrast to the conventional undercomplete representations, we have shown that our overcomplete network is competent to focus on high-level features such as edges and fine structures by constraining the receptive field of the network. Finally, the fused HSI is obtained by adding the residual HSI and the upsampled HSI. The comprehensive experiments conducted on three semisynthetic and one real HS datasets demonstrated the superiority of our DIP-HyperKite over the other state-of-the-art results in terms of qualitative and quantitative evaluations.

REFERENCES

[1] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[2] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[3] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[4] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[5] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[6] A. Mohammadzadeh, A. Tavakoli, and M. J. V. Zoej, "Road extraction based on fuzzy logic and mathematical morphology from pan-sharpened IKONOS images," *Photogramm. Rec.*, vol. 21, no. 113, pp. 44–60, Mar. 2006.

[7] G. Licciardi, A. Villa, M. M. Khan, and J. Chanussot, "Image fusion and spectral unmixing of hyperspectral images for spatial improvement of classification maps," in *Proc. Int. Geosci. Remote Sens. Symp.*, 2012, pp. 7290–7293.

[8] H. Zhang, H. Xu, X. Tian, J. Jiang, and J. Ma, "Image fusion meets deep learning: A survey and perspective," *Inf. Fusion*, vol. 76, pp. 323–336, Dec. 2021.

[9] G. Vivone *et al.*, "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[10] C. A. Laben and B. V. Brower, "Process for enhancing the spatial resolution of multispectral imagery using pan-sharpening," U.S. Patent 6 011 875, Jan. 4, 2000.

[11] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS + pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, May 2007.

[12] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[13] P. S. Chavez and A. Y. Kwarteng, "Extracting spectral contrast in Landsat thematic mapper image data using selective principal component analysis," *Photogramm. Eng. Remote Sens.*, vol. 55, no. 3, pp. 339–348, Mar. 1989.

[14] G. Vivone *et al.*, "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, Dec. 2014.

[15] T.-M. Tu, S.-C. Su, H.-C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, no. 3, pp. 177–186, Sep. 2001.

[16] S. G. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," in *Fundamantal Papers Wavelet Theory*. Princeton, NJ, USA: Princeton Univ. Press, 2009, pp. 494–513.

[17] G. P. Nason and B. W. Silverman, "The stationary wavelet transform and some statistical applications," in *Wavelets Statistics*. Springer, 1995, pp. 281–299. [Online]. Available: https://link.springer.com/chapter/10.1007/978-1-4612-2544-7_17

[18] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Dec. 2000.

[19] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[20] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[21] W. Liao, F. Van Coillie, S. Gautama, A. Pizurica, and W. Philips, "Fusion of thermal infrared hyperspectral and vis RGB data using guided filter and supervised fusion graph," Tech. Rep., 2011. [Online]. Available: https://telin.ugent.be/~wliao/IEEE_DataFusionContest2014_Liao.pdf

[22] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[23] Q. Wei, N. Dobigeon, and J. Y. Tourneret, "Bayesian fusion of multi-band images," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1117–1127, Sep. 2015.

[24] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J. Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[25] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[26] X. Tian, Y. Chen, C. Yang, and J. Ma, "Variational pansharpening by exploiting cartoon-texture similarities," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5400416.

[27] X. Fu, Z. Lin, Y. Huang, and X. Ding, "A variational pan-sharpening with local gradient constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10265–10274.

[28] C. Ballester, V. Caselles, L. Igual, J. Verdera, and B. Rougé, "A variational model for P+XS image fusion," *Int. J. Comput. Vis.*, vol. 69, no. 1, pp. 43–58, Aug. 2006.

[29] J. Ma, W. Yu, C. Chen, P. Liang, X. Guo, and J. Jiang, "Pan-GAN: An unsupervised pan-sharpening method for remote sensing image fusion," *Inf. Fusion*, vol. 62, pp. 110–120, Oct. 2020.

[30] Y. Zheng, J. Li, Y. Li, G. Jie, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

[31] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[32] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Oct. 2017, pp. 624–632.

[33] O. Sidorov and J. Y. Hardeberg, "Deep hyperspectral prior: Single-image denoising, inpainting, super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, 2019, pp. 3844–3851.

[34] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[35] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.

[36] J. Li *et al.*, "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.
[37] H. Xu, J. Ma, Z. Shao, H. Zhang, J. Jiang, and X. Guo, "SDPNet: A deep network for pan-sharpening with enhanced information representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4120–4134, May 2021.
[38] H. C. Burger, C. J. Schuler, and S. Harmeling, "Image denoising: Can plain neural networks compete with BM3D?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, May 2012, pp. 2392–2399.
[39] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 4681–4690.
[40] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Springer, 2015, pp. 234–241. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28
[41] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Jan. 2017.
[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Apr. 2016, pp. 770–778.
[43] M. S. Lewicki and T. J. Sejnowski, "Learning overcomplete representations," *Neural Comput.*, vol. 12, no. 2, pp. 337–365, 2000.
[44] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.
[45] J. M. J. Valanarasu, V. A. Sindagi, I. Hacihaliloglu, and V. M. Patel, "KiU-Net: Towards accurate segmentation of biomedical images using over-complete representations," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Springer, 2020, pp. 363–373. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-59719-1_36
[46] J. Maria Jose Valanarasu, V. A. Sindagi, I. Hacihaliloglu, and V. M. Patel, "KiU-Net: Overcomplete convolutional architectures for biomedical image and volumetric segmentation," 2020, *arXiv:2010.01663*.
[47] J. M. J. Valanarasu and V. M. Patel, "Overcomplete deep subspace clustering networks," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, May 2021, pp. 746–755.
[48] P. Guo, J. M. J. Valanarasu, P. Wang, J. Zhou, S. Jiang, and V. M. Patel, "Over-and-under complete convolutional RNN for MRI reconstruction," Tech. Rep., 2021. [Online]. Available: https://arxiv.org/abs/2106.08886
[49] S.-Y. Lo, J. Maria Jose Valanarasu, and V. M. Patel, "Overcomplete representations against adversarial videos," 2020, *arXiv:2012.04262*.
[50] R. Yasarla, J. M. J. Valanarasu, and V. M. Patel, "Exploring overcomplete representations for single image deraining using CNNs," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 2, pp. 229–239, Feb. 2021.
[51] K. Li, W. Xie, Q. Du, and Y. Li, "DDLPS: Detail-based deep Laplacian pansharpening for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8011–8025, Oct. 2019.
[52] Q. Wei, N. Dobigeon, and J. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.
[53] W. Xie, J. Lei, Y. Cui, Y. Li, and Q. Du, "Hyperspectral pansharpening with deep priors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1529–1543, May 2020.
[54] C. Harris and S. Mike, "A combined corner and edge detector," in *Proc. Alvey Vis. Conf.*, Manchester, U.K., vol. 15, no. 50, Sep. 1988, pp. 5210–5244.
[55] A. A. Mueller, A. Hausold, and P. Strobl, "HySens-DAIS/ROSIS imaging spectrometers at DLR," in *Proc. Remote Sens. Environ. Monitor.*, Jan. 2002, pp. 3–14.
[56] Y. Zeng, W. Huang, M. Liu, H. Zhang, and B. Zou, "Fusion of satellite images in urban area: Assessing the quality of resulting images," in *Proc. 18th Int. Conf. Geoinfor.*, Jun. 2010, pp. 1–4.
[57] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over chikusei," Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, 2016.
[58] Q. Wei, "Bayesian fusion of multi-band images: A powerful tool for super-resolution," Ph.D. dissertation, Inst. Nat. Polytechnique Toulouse, Toulouse, France, 2015. [Online]. Available: https://theses.eurasip.org/theses/651/bayesian-fusion-of-multi-band-images-a-powerful/
[59] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002.



Wele Gedara Chaminda Bandara (Student Member, IEEE) received the degree (Hons.) in electrical and electronic engineering from the University of Peradeniya, Peradeniya, Sri Lanka, in 2019. He is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Engineering (ECE), Johns Hopkins University, Baltimore, MD, USA.

He worked as a Graduate Research Student with the Department of Electrical and Electronic Engineering, University of Peradeniya, for an NSF-funded project from 2019 to 2020. His research interests include computer vision and image processing with applications in remote sensing and hyperspectral image processing.



Jeya Maria Jose Valanarasu (Student Member, IEEE) received the bachelor's degree in instrumentation and control engineering from the National Institute of Technology Tiruchirappalli, Tiruchirappalli, India, in 2019. He is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Engineering (ECE), Johns Hopkins University, Baltimore, MD, USA.

He also spent some time working with the Biomedical Engineering Department, National University of Singapore (NUS), Singapore, as a Visiting Research Intern. His research interests include image/3-D segmentation, image enhancement, and image-to-image translation for computer vision and medical imaging tasks.



Vishal M. Patel (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 2010.

He was an A. Walter Tyson Assistant Professor with the Department of Electrical and Computer Engineering, Rutgers University, and a Research Faculty Member with the University of Maryland Institute for Advanced Computer Studies (UMIACS), College Park. He is currently an Associate Professor with the Department of Electrical and Computer Engineering (ECE), Johns Hopkins University, Baltimore, MD, USA.

Dr. Patel has received a number of awards, including the 2021 IEEE Signal Processing Society (SPS) Pierre-Simon Laplace Early Career Technical Achievement Award, the 2021 NSF CAREER Award, the 2021 IAPR Young Biometrics Investigator Award (YBIA), the 2016 ONR Young Investigator Award, the 2016 Jimmy Lin Award for Invention, the A. Walter Tyson Assistant Professorship Award, Best Paper Awards at IEEE AVSS 2017 and 2019, IEEE BTAS 2015, IAPR ICB 2018, and IEEE ICIP 2021, and two Best Student Paper Awards at IAPR ICPR 2018. He is an Associate Editor of the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and *Pattern Recognition Journal*. He serves on the Machine Learning for Signal Processing (MLSP) Committee of the IEEE Signal Processing Society. He serves as the Vice President of conferences for the IEEE Biometrics Council.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:57:08 UTC from IEEE Xplore. Restrictions apply.